PeriGen®

Empowering Clinicians. Advancing Perinatal Systems.

Saint Luke's Health Systems PeriCALM Tracings Preliminary System Architecture

Wright and Hedrick Activation

Vitals and Annotations to Epic

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1 Purpose and Use

This document is meant to provide the client with preliminary understanding of the hardware requirements associated with the additional site activations (Wright and Hedrick) as well as the implementation of Vitals and Annotations to Epic.

Sections covered in the document are:

1. Purpose and Use
2. Assumptions used to build the configuration
3. System Architecture
4. Closet Architecture
5. Hardware Sizing Specifics
6. Supporting Information

Please note that this document does not reflect the final configuration. The information in this document is based on the current information available as part of the pre-sales process. Complete configuration, diagram and precise technical specification will be provided as part of pre-implementation activities and comprehensive site survey.

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2 Assumptions

The below assumptions were used while building the hardware requirements based on preliminary reviews.

- The hardware implemented at the five active sites (Cushing, Plaza, Northland, Lee's Summit, and South) are compliant with the PeriCALM Tracings 3.18 requirements
- The deployed hardware model with redundancy at the Database and Acquisition servers implemented at the five sites has been mirrored for the Wright and Hedrick sites.
- Each of the Wright and Hedrick sites will have a single closet to host the Surveillance Acquisition servers.
- A stable hard-wired network is in place to accommodate all servers and workstation requirements (100 Mbit switched with under 20ms latency or better).
- Dedicated Cat5/6 runs with RJ45 terminations are required for attaching fetal monitors in each room or area required for monitoring. These lines should be fully compatible with RS-232 specifications. Direct, point-to-point cabling is strongly recommended.
- For cart based architecture housing both fetal monitor and computer, 2 RJ45 wall plates are required. One for the fetal monitor (previous bullet) and one for the workstation to connect to the hospital network
- PeriGen will provide proprietary cables between each fetal monitor and the wall-plate.
- The network closet in each ward is secure, environmentally controlled and appropriate for the installation of required hardware.
- All non-OS partition storage is fault protected via RAID5 or RAID 10 preferably on an enterprise class storage array.

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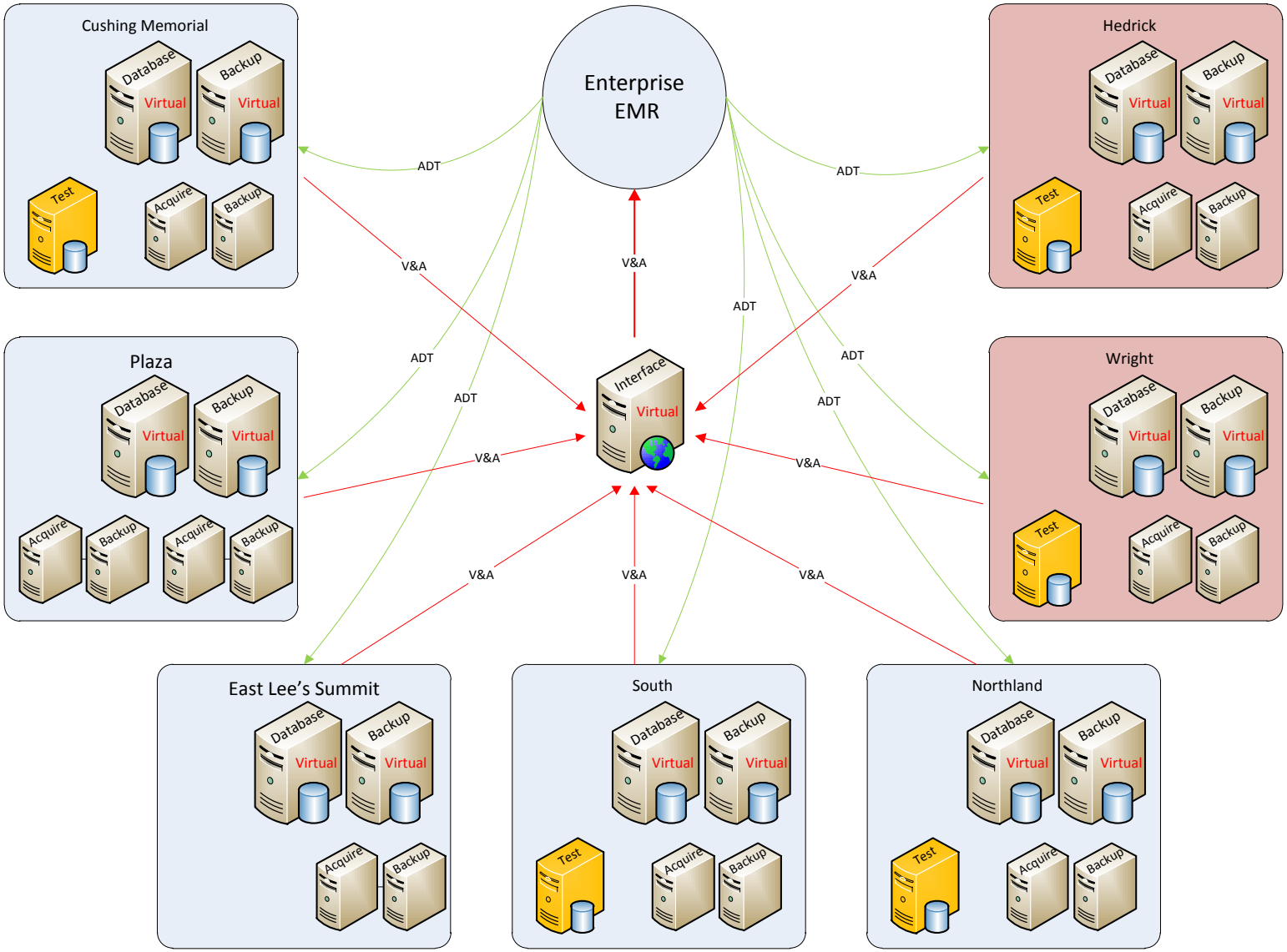
3 System Architecture



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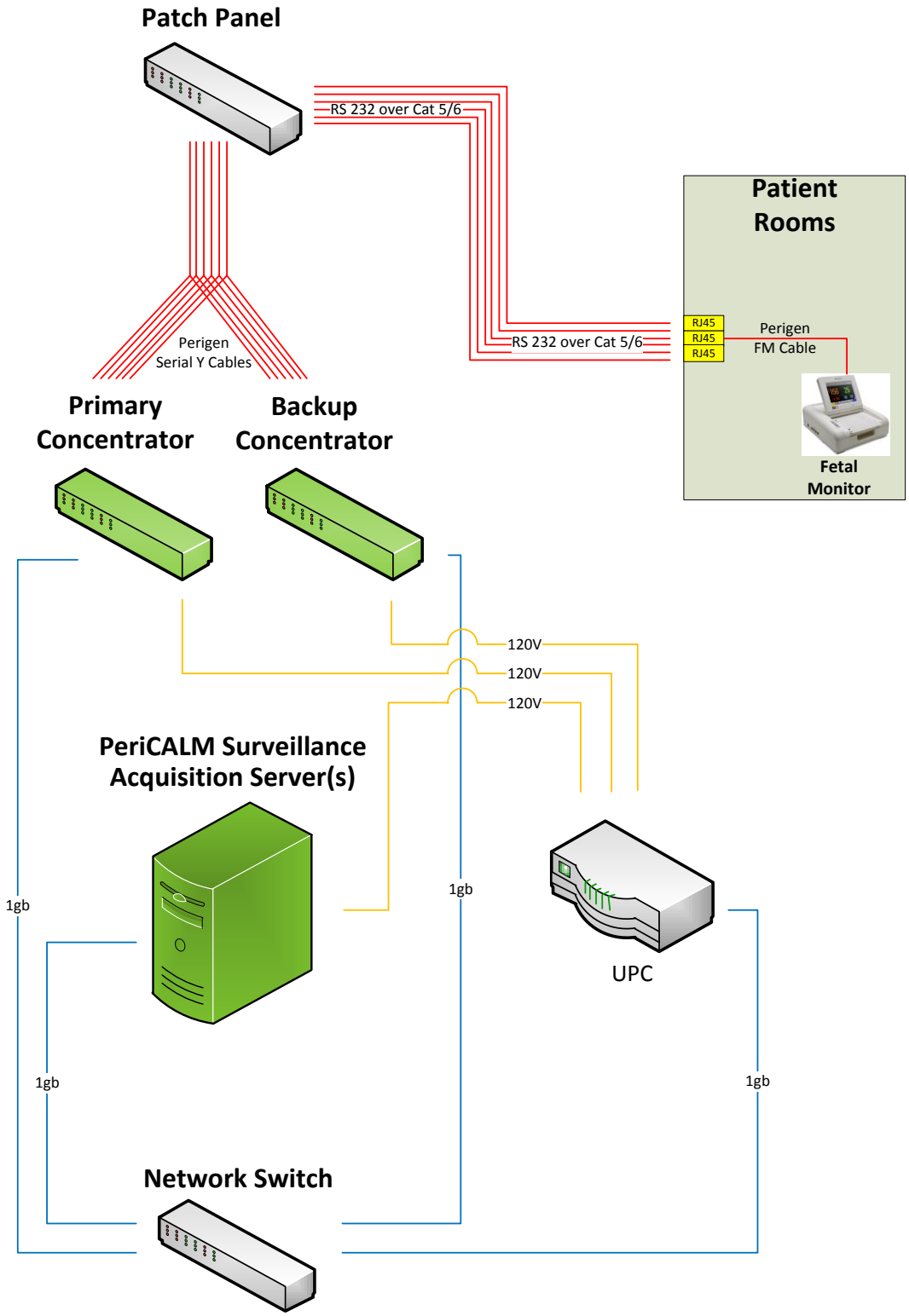
4 Closet Architecture



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6 Site

Specific Hardware/Software Checklist

The below checklist is to provide the client with a summary list of additional hardware needs.

Hedrick server requirements:

Quantity	Function	RAM	Processors	Storage	Required Licenses
2	PeriCALM Tracings Primary Server (Virtual)	4gb	1x Intel Quad Core E5620*	2x72gb RAID 1 Arrays	Windows 2008 R2 Standard Edition
2	PeriCALM Surveillance Acquisition Server	4gb	1x Intel Dual Core*	2x72gb RAID 1 Arrays	Windows 2008 R2 Standard Edition
1	Test Server - All modules (Virtual)	4gb	1x Intel Quad Core E5620*	2x72gb RAID 1 Arrays	Windows 2008 R2 Standard Edition

*Or Better

(2) Surveillance Concentrators per closet. (See certified models in section 7)

(1) UPS Devices per closet

(1) PeriGen Serial Y Cables per FM Port. (Supplied by PeriGen)

(1) PeriGen FM Cables per Fetal Monitor. (Supplied by PeriGen)

Wright server requirements:

Quantity	Function	RAM	Processors	Storage	Required Licenses
2	PeriCALM Tracings Primary Server (Virtual)	4gb	1x Intel Quad Core E5620*	2x72gb RAID 1 Arrays	Windows 2008 R2 Standard Edition
2	PeriCALM Surveillance Acquisition Server	4gb	1x Intel Dual Core*	2x72gb RAID 1 Arrays	Windows 2008 R2 Standard Edition
1	Test Server - All modules (Virtual)	4gb	1x Intel Quad Core E5620*	2x72gb RAID 1 Arrays	Windows 2008 R2 Standard Edition

*Or Better

(2) Surveillance Concentrators per closet. (See certified models in section 7)

(1) UPS Devices per closet

(1) PeriGen Serial Y Cables per FM Port. (Supplied by PeriGen)

(1) PeriGen FM Cables per Fetal Monitor. (Supplied by PeriGen)

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Vitals and Annotations Interface server requirement:

Quantity	Function	RAM	Processors	Storage	Required Licenses
1	Production Interface Server (Virtual)	4gb	1x Intel Quad Core E5620*	2x72gb RAID 1 Arrays	Windows 2008 R2 Standard Edition
1	Test Interface Server (Virtual)	2gb	1x Intel Dual Core*	2x72gb RAID 1 Arrays	Windows 2008 R2 Standard Edition

*Or Better

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7 Supporting Information

Client / Workstation Requirements

- Microsoft Windows XP Professional 32-bit (SP3) or Microsoft Windows 7 Professional 32/64-bit
- 2GB RAM, 4GB recommended. 1GB of available memory should be reserved for all PeriCALM Tracings modules.
- Internet Explorer 7.0, 8.0 or 9.0. Ensure that client policies do not restrict the installation of trusted ActiveX controls by end users.
- 1024x768 or 1280x1024 (4:3) is the standard minimal resolution supported by the system. Wide screens in Portrait mode (1680x1050 or 1920x1200) are recommended for optimal user experience. Only native resolution and native aspect ratios are supported (non-native resolutions or aspect ratios will result in a stretched representation of fetal tracings and are not supported)
- Citrix deployment is supported on dedicated servers only.

Supported Tracing acquisition concentrators per networking closet

- DIGI Etherlite EL-32, 32 port (part # 70001429).
- DIGI Etherlite EL-160, 16 port (part # 70001427).
- MOXA nPort 5610. 16 port (part # NPort 5610-16).

  The total number of devices is determined by total number of monitoring areas and distribution across networking closets. Redundant configuration requires double the number base devices.

Network

- Switched Ethernet network, 100 Mbit or better
- Guaranteed maximum latency of 20ms (worse case) between all components of the system
- CAT5+ dedicated cabling between each monitoring area and networking closet. Cables should be compliant to RS232 data transmission standards. Point-to-point configuration recommended.

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Virtual Environment Support

- Virtual servers on VMware VSphere v4.X and v5.1 are supported on equivalent or better server specifications. Other platforms may require extra validation services.
- Resource reservations are mandatory on virtual servers. Certain restrictions may be needed depending on site layout and load.
- All Servers are supported in a virtual environment but it is highly recommended that the Surveillance acquisition Server not be virtualized due to potential Time Synchronization issues.

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